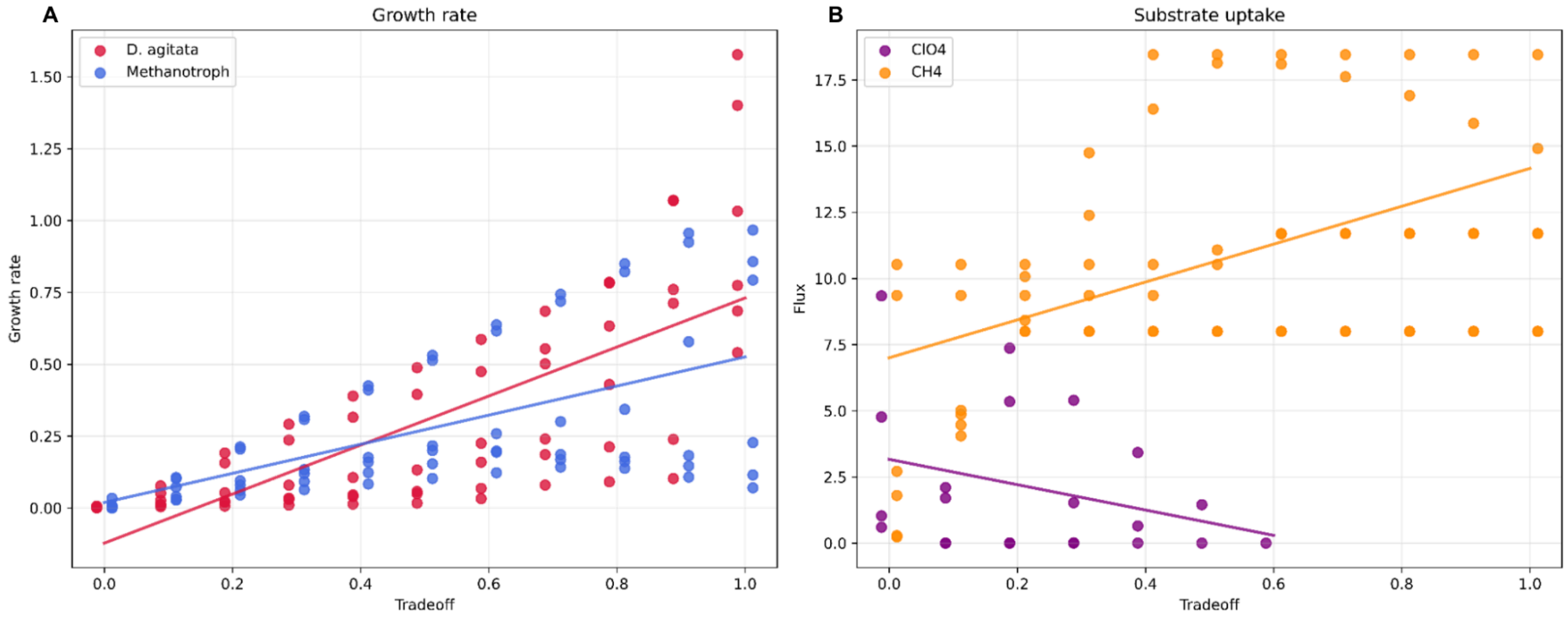
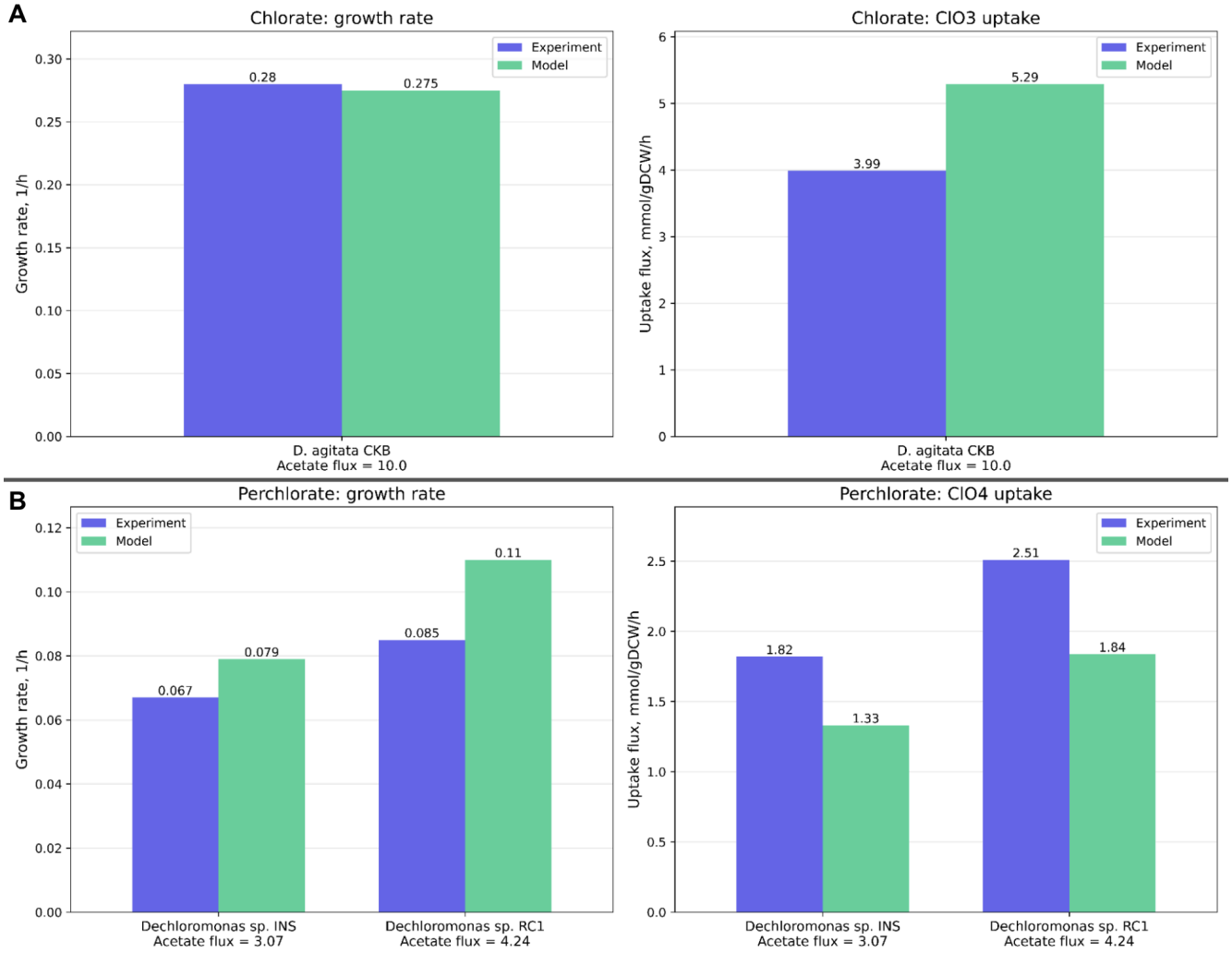
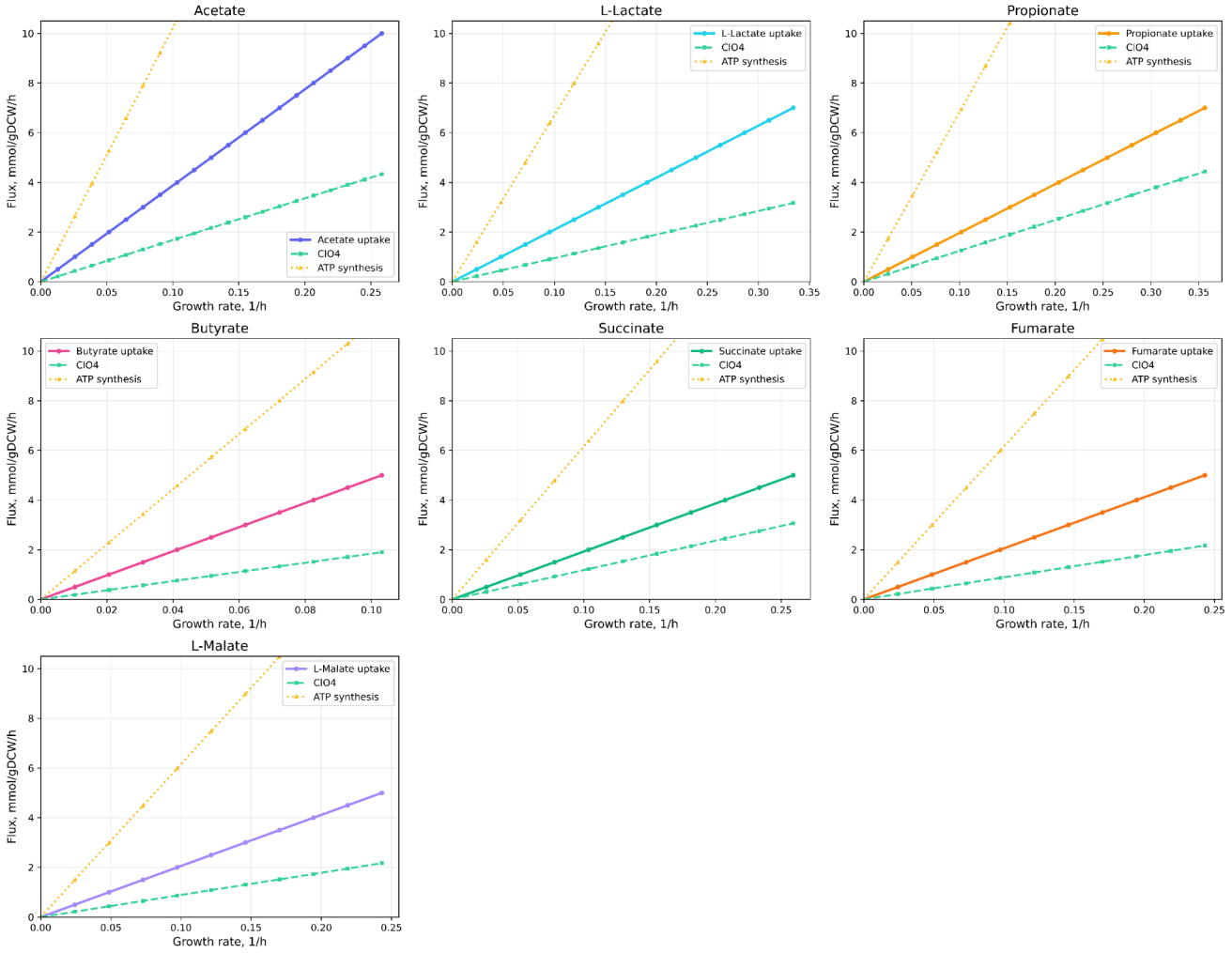


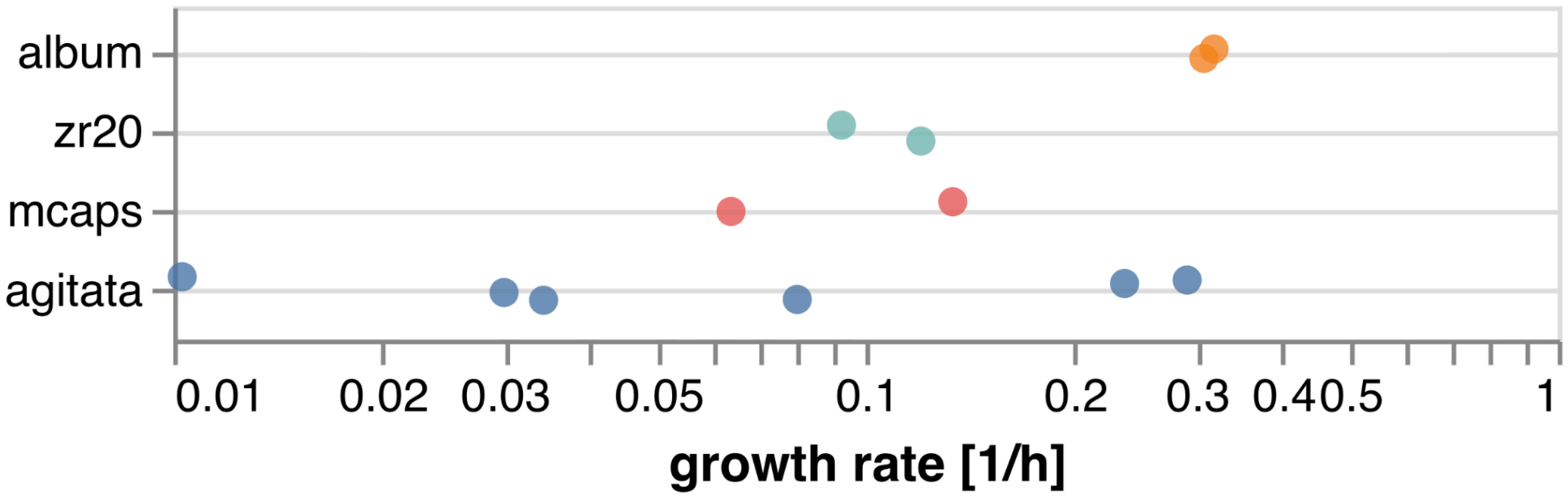
Supplementary Figure 1. Effects of the MICOM tradeoff parameter on community member growth rates (A) and methane and perchlorate uptake rates (B)



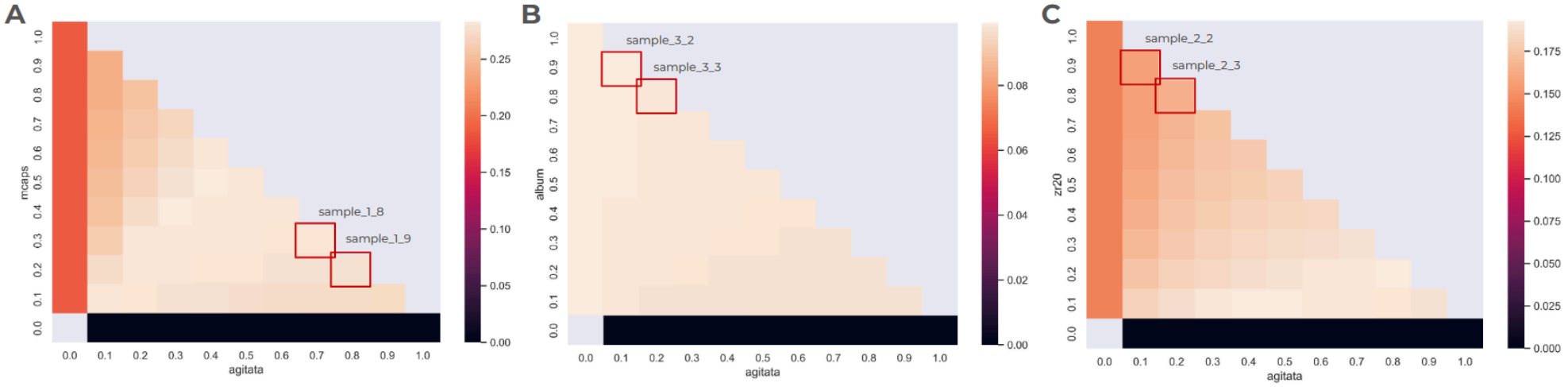
Supplementary Figure 2. Comparison of experimentally measured and model-predicted growth rates and electron acceptor uptake rates for *Dechloromonas* strains. (A) *D. agitata* CKB grown with chlorate. (B) Strains INS and RC1 grown with perchlorate



Supplementary Figure 3. Analysis of model growth under different substrates: acetate, lactate, propionate, butyrate, succinate, fumarate, malate



Supplementary Figure 4. Community models reconstructed using MICOM and exhibiting non-zero growth rates



Supplementary Figure 5. Community models reconstructed by PyCoMo. Ratios corresponding to MICOM communities are highlighted in red